NURSING CARE OF A FAMILY WHEN A CHILD HAS AN UNINTENTIONAL INJURY

Chapter 52

OBJECTIVES

- Assess a child with unintentional injuries.
- Formulate nursing diagnoses related to an unintentionally injured child.
- Establish expected outcomes for an unintentionally injured child that can help parents/caregivers manage seamless transitions across differing healthcare settings.
- Describe the causes and consequences of common unintentional injuries in childhood and measures to prevent them.
- Evaluate expected outcomes for achievement and effectiveness of care.
- Integrate knowledge of unintentional injury in children with the interplay of the nursing process, the six competencies of QSEN, and family nursing to achieve quality maternal and child health nursing care.

OVERVIEW

- According to the Centers for Disease Control and Prevention Childhood Injury Report (CDC, 2021a), the leading cause of morbidity and mortality among children in the United States is unintentional injuries caused by burns, drowning, falls, poisoning, and motor vehicle accidents.
- More than 12,000 children die and 9.2 million are treated in emergency departments due to unintentional injuries yearly in the United States. For this reason, all healthcare providers must be aware of how to prevent childhood injuries and care for children with injuries

ASSESSMENT

- When children are seen for unintentional injury, parents and guardians are probably not functioning at 100% due to stress
- May be frightened not only by what happened but also what could have happened
- Children often feel guilty and worry they will be punished
- Their parents can feel equally guilty because they realize they could have been supervising more closely, causing them to act defensively
- Remember people under stress do not process information well, so parents may not perceive the information you give them correctly
- All information you supply in an emergency department, therefore, may need to be repeated after the immediate care is completed.

NURSING DIAGNOSIS

- Nursing diagnoses need to address not only the physical injury a child sustained but also the psychological or emotional trauma both the child and the parents are experiencing.
- Examples of possible nursing diagnoses are:
  - Pain related to fractured tibia from sports injury
  - Ineffective airway clearance related to effect of burned esophageal tissue
  - Impaired physical mobility related to severe burn injury
  - Altered body image perception related to change in physical appearance from thermal injury
  - Parental fear related to outcome after head injury in child Impaired family processes related to child's unintentional injury
  - Anxiety related to apprehension and lack of knowledge regarding medical treatment of child

OUTCOME IDENTIFICATION AND PLANNING

- Parents in an emergency department are rarely ready for long-term planning; they often have great difficulty in just answering the most straightforward, immediate questions
- At time of discharge from the emergency department, be certain that parents/caregivers have
  - Written and verbal instructions about the child's care
  - Name and number of the person to contact if they have questions about care or progress
  - An appointment (or the number to call for a return appointment) for follow-up care
  - A key component of nursing intervention in an emergency department is to help parents understand why an injury happened and plan ways to make their immediate or community environment safe for children

IMPLEMENTATION

- The extent of a child's injury depends on the injuring agent, the site that was injured, and often the immediate care, including both physical and psychological management that a child receives.
- Be certain to maintain standard infection precautions in emergencies the same as at any other time.
- Parental consent must be obtained for interventions, except for lifesaving actions, such as cardiopulmonary resuscitation (CPR).
    - It is assumed that parents would consent to lifesaving procedures, and delaying procedures until they can be located could result in permanent disability or death

OUTCOME EVALUATION

- After an injury, children need follow-up care to monitor the progress. At the follow-up visit, an assessment of both parental/caregiver psychological reaction and the child's physical healing and psychological adjustment to the injury can be obtained
- Examples of expected outcomes suggesting achievement of goals are:
  - The child is able to ingest fluids without distress after esophageal burns.
  - The child does not display evidence of pain (age dependent) within 20 minutes of administration of pain medication.
  - The child demonstrates full range of motion in hand after thermal injury.
  - The child states understanding (age dependent) that wearing a seat belt is an important safety measure.
  - The child affirms (age dependent) wearing a helmet when riding a bicycle in the future.

HEALTH PROMOTION AND RISK MANAGEMENT

- In a variety of settings, nurses have the opportunity, as healthcare providers, to provide education to children and families in the prevention of unintentional injuries.
- After an injury has occurred, nurses provide valuable anticipatory guidance to families about injury prevention.
- This chapter presents anticipatory guidance information and age-appropriate injury prevention after each section.

THE INJURED CHILD

- Nurse in ED may be first healthcare provider to care for child after injury
- Obtains initial assessment
  - May be in shock and not display emotions
  - May have internal injuries and not be able to verbalize how that feels
- Injury becomes life threatening when vital organs affected
- Evaluation of airway, breathing, circulation, and disability (A, B,C,D) is done in this order
- Table 52.1 lists signs and symptoms to assess when determining status of child

THE INJURED CHILD

- Conduct brief history
- Can be difficult to evaluate child because they may be unconscious, too young to communicate, or emotional
- Provide frequent explanations to guardians and permit them to be in the room with the child – can be helpful to have someone on team assigned to this role
- A proportion of unintentional injuries are related to child maltreatment
    - If suspected a report must be made

HEAD TRAUMA

- Common childhood injury- most times minor
- Falls are leading cause of head trauma, followed by unintentional blunt trauma, and thirdly MVA
- Always potentially serious and can lead to subsequent neuro complications

HEAD TRAUMA – ANTICIPATORY GUIDANCE

- Children should ride a properly fitted bicycle, wear a properly fitted helmet, and ride in a safe environment that is appropriate for their age.
- Parents/caregivers should supervise children (age appropriate) during play and sports.
- Parents/caregivers should purchase, install, and correctly use car seats/booster seats (NHTSA, 2021).
- Parents/caregivers should ensure the home and play area of their children has appropriate injury prevention devices.
- Resources include the National Highway Transportation Safety Administration and the American Academy of Pediatrics.

HEAD TRAUMA – IMMEDIATE ASSESSMENT/ MANAGEMENT

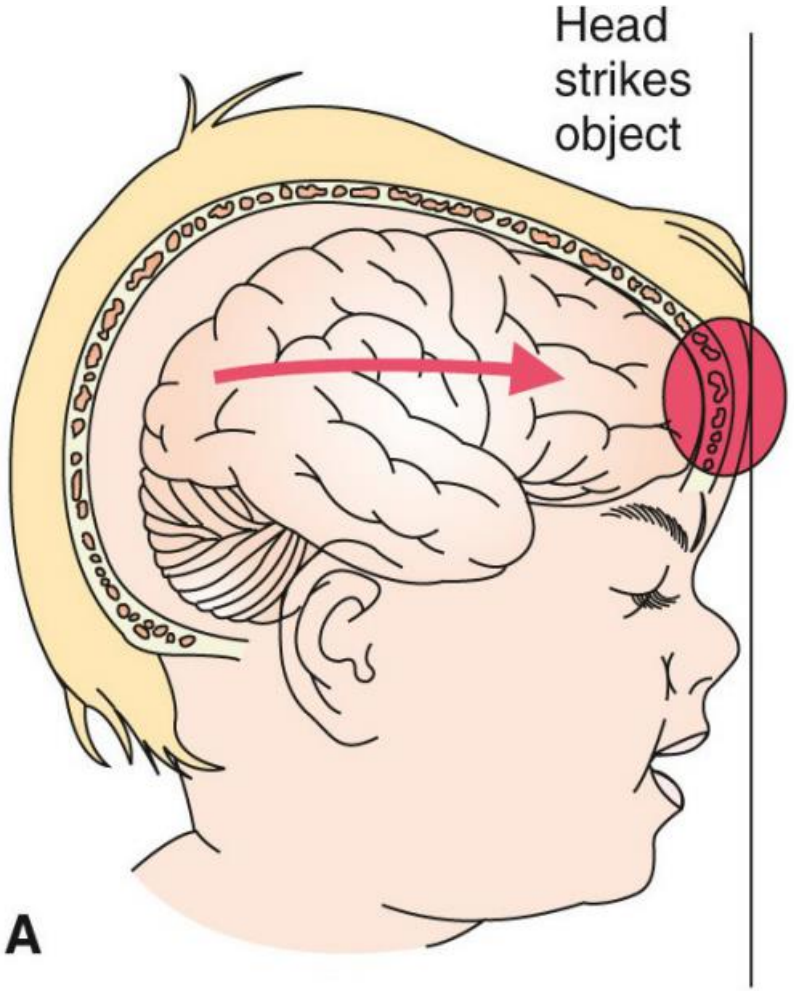
- 1ST- All children with head trauma need to have their head and neck stabilized with a backboard and cervical collar until cervical injury has been ruled out.
- They also require immediate and 2ND- continual neurologic assessment to identify any decline in neurologic functioning
- After a head injury, brain edema is likely to occur - ICP monitoring should be initiated
- A CT scan or MRI will determine areas of edema or bleeding.
- A central venous or central arterial line may be inserted, so pressure can be reduced with the administration of a hypertonic intravenous (IV) solution such as mannitol.
- Administration of a steroid such as dexamethasone and keeping the child's head elevated are other ways to decrease inflammation and edema and lower ICP

SKULL FRACTURE

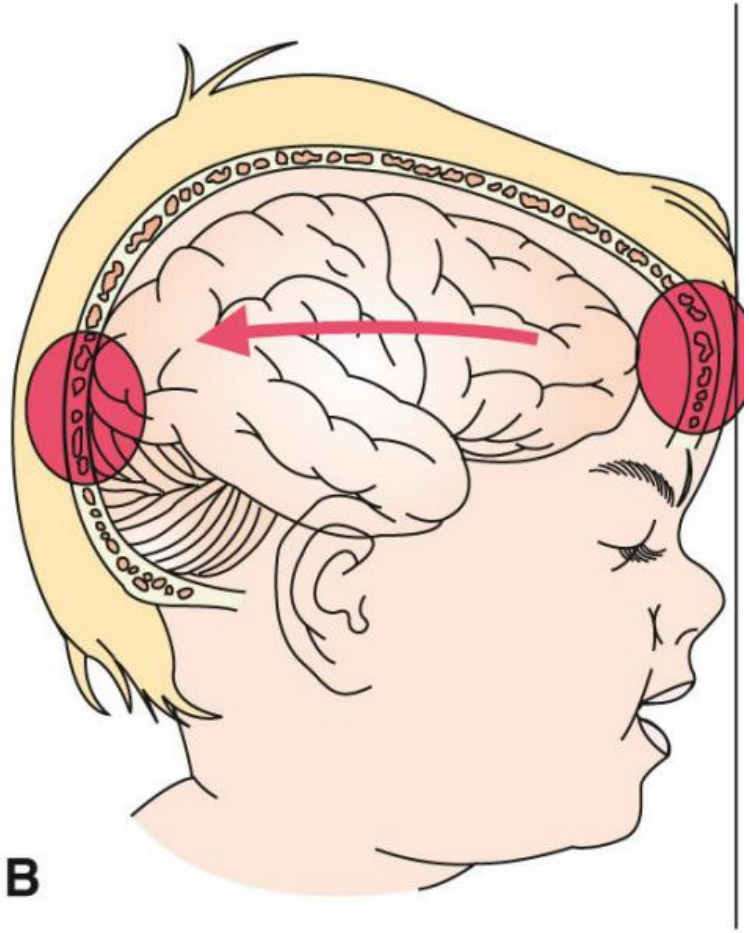
- A skull fracture causes a defect in the bone of the skull
- Confirmed by skull x-ray
- It is important to obtain a thorough history of the injury so that the mechanism of injury can be determined
- If the base of the skull is fractured (basal skull fracture), a child usually exhibits ecchymosis around the eyes (raccoon eyes) or behind an ear.
  - Rhinorrhea or otorrhea may be present (CSF) - test fluid with glucose reagant strip and it will be positive
- Keep child in semi-fowlers position
- May be prescribed prophylactic antibiotic
- **NEVER PLACE AN NG TUBE IN PATIENT WITH SUSPECTED SKULL FRACTURE**

CONCUSSION

- Concussion is defined as a brain injury induced by biomechanical forces transmitted to the head resulting in rapid onset of transiently impaired neurologic function
- Children have at least a transient loss of consciousness at the time of the injury and may vomit and show irritability after regaining consciousness
- Skull x-ray or MRI will rule out skull fracture
- Observation for 24 hours is needed
- Can be observed at home if parents can assess LOC every 1-2 hours while awake
  - Educated to not wake child multiple times during the night – can be disorienting
  - Instruct them to wake child at least once to determine LOC
  - Ensure they have number to call if they have questions



COUP INJURY
Anterior of brain strikes skull and is injured



CONTRECOUP INJURY
Brain recoils and strikes posterior skull, so is injured twice

ABDOMINAL TRAUMA

- Abdominal trauma is present in approximately 25% of pediatric patients with major trauma and is the most common cause of unrecognized fatal injury in children
- Abdominal trauma can result from a blunt object to the abdomen, a motor vehicle accident with or without restraint, child maltreatment, or a fall
- Injury to the spleen is more common in children than adults, who usually sustain injury to the liver
- The injury may be difficult to detect, however, because, unlike obvious signs of head or respiratory injuries, abdominal trauma usually presents with much more subtle signs
- Knowledge of the mechanism of injury can aid in the diagnosis

ABDOMINAL TRAUMA ASSESSMENT

- To detect abdominal trauma, assess vital signs for a baseline and reassess to monitor for any changes
- Abdominal hemorrhage is exhibited by rapid respirations, hypotension (less than 80 mm Hg systolic pressure in an older child; less than 60 mm Hg in an infant), and increasing pallor
- If low blood pressure results in little improvement when IV fluid is administered, internal bleeding may be the underlying factor
- If suspected, NG tube and foley can be placed & monitored for blood
- CT is gold standard
- Parents/caregivers may not seek immediate medical attention after abdominal trauma because they are unaware that a serious injury has resulted

SPLENIC INJURY

- In children, the spleen is the most frequently injured organ when there is abdominal trauma because of its location under the lower left ribs
- Mechanisms of injury involving the spleen are improper use of seat belts, handlebar injuries in bicycle accidents, or skateboard or snowboard accidents
- child will have tenderness in the left upper quadrant of the abdomen, especially on deep inspiration, when the diaphragm moves down and touches the spleen
- To keep this from happening, the child may hold the left shoulder elevated, so that the diaphragm is raised on the left side, or report radiated left shoulder pain while lying in a supine position (Kehr sign)
- Children with suspected splenic injury are usually admitted to an observation unit if the blood loss from rupture appears to be mild. If bleeding is severe, immediate surgery, such as a partial or total splenectomy, may be necessary to halt the bleeding and save the child's life. After a splenectomy, children are very susceptible to infection

ABDOMINAL TRAUMA – ANTICIPATORY GUIDANCE

- Using guards on windows, stair gates, and guard rails to prevent children from dangerous falls
- Ensuring playground equipment is well always maintained and providing supervision for young children when they are out at play
- Proper installation and use of car seats, booster seats, and seat belts
- Parental/caregiver modeling in their use of seat belts

DROWNING

- Drowning is death caused by suffocation from submersion in liquid when the inhaled water fills the lungs and therefore blocks the exchange of oxygen in the alveoli
- More than 3,500 children diet from drowning every year
- When a child is pulled from the water after drowning, mouth-to-mouth resuscitation and CPR should be started immediately
- Anticipatory guidance
  - Use fencing and gating of appropriate height and material around swimming pools.
  - Ensure adequate and consistent adult supervision of children during water activities.
  - Use approved flotation devices while swimming or boating.
  - Toddlers can drown in shallow areas, including toilets and buckets, and therefore should always be adequately supervised

POISONING

- When providing patient education, poison prevention information for parents should include:
  - Keep medicines and toxic products locked and out of reach of children.
  - Include poison control number (1-800-222-1222) in contact list.
  - Call the poison control center if you think a child has ingested anything that may be poisonous.
    - The child may be awake, alert, and acting normally. If a child has ingested a poisonous product and collapses or stops breathing, call 911.
  - When administering medications, be sure to read labels carefully and to administer them according to instructions.
  - Safely dispose of unused, unneeded, or expired medications and vitamins

EMERGENCY MANAGEMENT OF POISONING

- the best method to deactivate a swallowed poison is the administration of activated charcoal, either orally or by way of an NG tube
- Can have lead poisoning, iron poisoning, acetaminophen poisoning, or causitic (MAY BE VERY IRRITATING WHEN THEY VOMIT)
- Activated charcoal is supplied as a fine black powder that is mixed with water for administration.
  - Adding a sweet syrup to the mixture can make it more palatable.
  - Caution parents that, as the charcoal is excreted through the bowel over the next 3 days, stools will appear black

ACTIVATED CHARCOAL

- Action: absorbs toxic substances that have been swallowed to prevent them from being absorbed by the stomach
- Dosage : provided as a powder that must be mixed with water and administered orally or by way of nasogastric (NG) tube
- Possible adverse reactions: vomiting, diarrhea, black stools
- Nursing implications:
  - Administer orally to conscious victims only.
  - Give the drug as soon as possible after poisoning.
  - Store the drug in a closed container because if it absorbs gases from the air, it is inactivated.
  - Recognize that the solution feels gritty and tastes disagreeable so young children often have difficulty swallowing it, which is why it may be administered by NG tube.
  - Caution child and parents that stools will be black for several days after administration.

FOREIGN BODY OBSTRUCTION

- Foreign bodies can become lodged in any orifice of a child, such as the esophagus, ear canals, nostrils, or lungs.
- If the object is in the lungs, it can cause direct obstruction of the airway or long-term stasis of secretions and infection.
- Whether a foreign substance is inhaled or embedded elsewhere, nursing interventions should focus first on comforting the child and aiding in removal of the object and then focus on educating the parents/caregivers on prevention of foreign body obstructions

FOREIGN BODY IN ESOPHAGUS OR STOMACH

- Hard candy, coins, button batteries, small toys, magnets, and orthodontic appliances are common objects that may be lodged in the esophagus
- A symptom of an object lodged in the esophagus is intense pain at the site where the object is located.
- Magnets, such as those found in watches or hearing aids, need to be removed by endoscopy
- This should be attempted as soon as possible because they can lead to bowel perforation or volvulus from the acid content in the magnet
- Objects, such as a part of a toy, a chicken bone, or a large coin that will not dissolve or be passed, are removed by endoscopy
- Small coins, such as pennies and dimes, may pass by through the GI tract without difficulty.
- Parents/caregivers and older children need to observe stools over several days to determine that the coin did pass through the GI tract (about 48 hours after ingestion). Inform parents to observe for signs of bowel perforation or obstruction, such as vomiting or abdominal pain, until the coin has passed. After 3 to 7 days, if the child is symptom free, an X-ray may be taken to locate the coin

SNAKEBITE

- Immediately after a pit viper bite, a white wheal surrounding the puncture marks begins to appear, accompanied by excruciating pain at the site.
- Purplish erythema and edema extend rapidly from the site.
- By the time a child is seen at a healthcare facility, sanguineous fluid may be oozing from the bite.
- Systemic symptoms, such as dizziness, vomiting, perspiration, and weakness, may be present.
- Because snake venom interferes with blood coagulation, the child may have hematemesis or bleeding from the nose, intestines, or bladder because of subcutaneous or internal hemorrhage.
- The pupils may be dilated, showing the potent effect on cerebral centers.
- If the envenomation is not treated, seizures, coma, and death may result

SNAKEBITE-EMERGENCY MANAGEMENT AT SCENE

- Parents should apply a cold compress to the bite because this can decrease the spread of the venom and reduce edema formation.
- Lay the child to lie quietly, with the bitten extremity dependent, also helps to slow circulation.
- Commercial snakebite kits have rubber suction cups in them that can be used to suction out the venom, so if available, they should be used.
- Excising the bite with a knife and sucking out the venom is not recommended.
- It is vital to transport the child to a healthcare facility as quickly as possible.

SNAKEBITE – MANAGEMENT AT FACILITY

- In the emergency department, while obtaining a history, try to get a description of the snake
- Antivenin may contain a horse-serum base.
  - Before the serum is injected intramuscularly or intravenously, a skin test may need to be performed to prevent a possible anaphylactic reaction to the horse serum.
- If giving the serum intramuscularly, do not inject it into an edematous body part because medication will be poorly absorbed from edematous tissue.
- If the child's immunization status is unknown or it has been more than 5 years since a tetanus immunization was given, a booster may be administered.

ATHLETIC INJURIES

- When providing patient education, sports injury prevention information for parents/caregivers should include
  - Ensure children wear protective equipment that fits appropriately and correctly at all times during sports and recreational activities.
  - Ensure children are well hydrated with water (avoid sugary drinks) and appropriately dressed to avoid heat-related injuries or illness.
  - Parents should be good role models and wear helmets and protective equipment when participating in sports and recreational activities

FROSTBITE

- Frostbite is tissue injury caused by cold exposure. The extreme cold causes peripheral vasoconstriction, limiting the oxygen supply to surrounding cells
- The severity of frostbite is classified by degree:
  - First degree: mild freezing of epidermis; appears erythematous with edema
  - Second degree: partial- or full-thickness injury; appears erythematous; blisters and pain occur after rewarming
  - Third degree: full-thickness injury (epidermis, dermis, and subcutaneous tissue); appears white
  - Fourth degree: complete necrosis with gangrene and possible ultimate loss of body part

FROSTBITE MANAGEMENT

- Warm frostbitten areas gradually because sudden warming increases the metabolic rate of cells, which can cause additional damage.
- Administration of a vasodilator and use of hyperbaric oxygen may help reduce the effect on body cells
- When frostbitten body parts begin to warm – very painful so good pain management is imperative
- Severe frostbite - necrosis of destroyed tissue will occur and slough off. Apply or change dressings PRN to prevent secondary infection

BURNS

- Burns are injuries to body tissue caused by excessive heat (greater than 104°F [40°C]).
- Burns are the second most common unintentional injury seen in children 1 to 4 years of age and the third most common cause of injury in children 5 to 14 years of age.
- Younger children are most at risk for scald burns that are caused by hot liquids or steam.
- Older children are more likely to be burned from flames after they move too close to a campfire, heater, or fireplace; touch a hot curling iron; or play with matches or lighted candles
- Always assess for child maltreatment

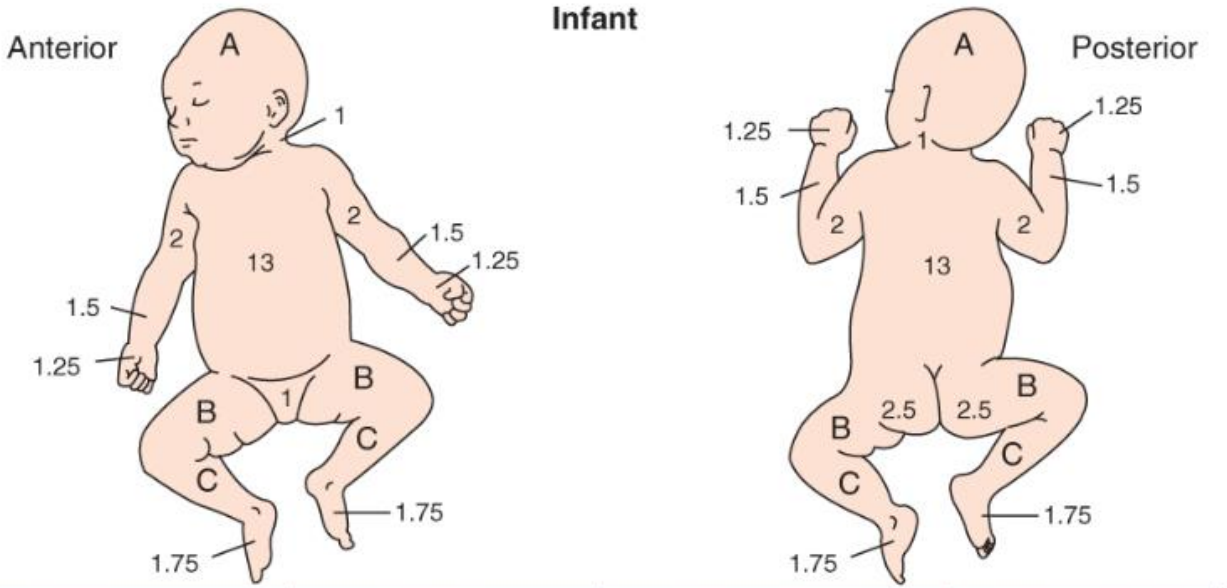
BURNS

- Burn injuries tend to be more serious in children than in adults because the same-sized burn covers a larger surface of a child's body
- Many burns can be prevented with parent/caregiver and child education
- When providing patient education, burn prevention information for parents/caregivers includes:
  - Install smoke alarms in the home, on every floor, and near all rooms in which family members sleep.
  - Monitor smoke alarms by testing them monthly to make sure they are working properly.
  - Use long-life batteries.
  - Create a family fire escape plan, involve the children in the planning, and practice frequently.
  - Never leave food unattended on the stove and always supervise or restrict the use of stoves, ovens, and microwaves with children. (TURN COOKWARE HANDLE TOWARDS BACK AND USE BACK STOVE)
  - Check the water heater temperature and make sure that the thermostat is set to 120°F or lower. Always test the water before children enter the bathtub or shower

BURNS ASSESSMENT

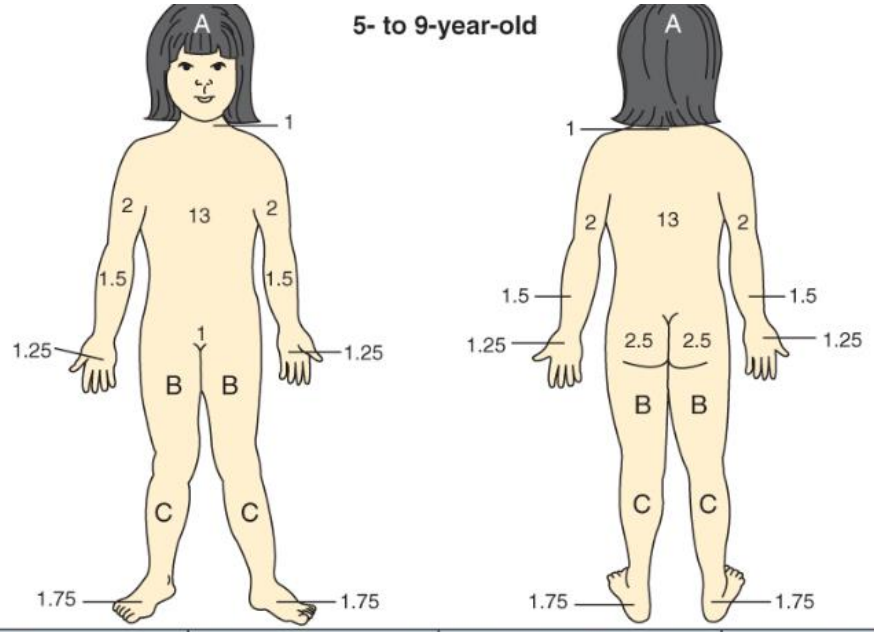
- Face and throat burns are hazardous because there may be accompanying but unseen burns in the respiratory tract that could lead to respiratory tract obstruction
- Hand burns are hazardous because if the fingers and thumb are not positioned properly during healing, adhesions will inhibit full range of motion in the future
- Burns of the feet carry a high risk of secondary infection
- Genital burns are also hazardous because edema of the urinary meatus may prevent a child from voiding
- Rule of nines is used for adults but can be misleading in children – next semester

BURNS ASSESSMENT



AREA	BIRTH	AGE 1 YEAR	AGE 5 YEARS
A = 1/2 of head	9 1/2	8 1/2	6 1/2
B = 1/2 of one thigh	2 3/4	3 1/4	4
C = 1/2 of one leg	2 1/2	2 1/2	2 3/4

BURNS ASSESSMENT



AREA	AGE 10 YEARS	AGE 15 YEARS	ADULT
A = 1/2 of head	5 1/2	4 1/2	3 1/2
B = 1/2 of one thigh	4 1/2	4 1/2	4 3/4
C = 1/2 of one leg	3	3 1/4	3 1/2

EMERGENCY MANAGEMENT FOR MINOR BURNS

- All burns, including minor burns, need immediate care because of the potential pain involved
- Although minor burns (typically first-degree partial-thickness burns) are the simplest type of burn, they involve pain and death of skin cells and therefore must be treated seriously
- Immediately apply cool water to decrease the skin temperature and prevent further burning
- Application of an analgesic–antibiotic ointment and a gauze bandage to prevent infection is usually the only additional treatment required
- Be certain that parents have a follow-up appointment in about 2 days to have the dressing changed and the area inspected for a secondary infection
- Advise parents/caregivers to keep the dressing dry (no swimming or getting the area wet while bathing) until the burn is healed—about 1 week

EMERGENCY MANAGEMENT FOR MODERATE BURNS

- Moderate or second-degree burns typically are blistered
- It is not necessary to drain the blisters initially, as this is a source of infection
- The burn should be covered with a topical antibiotic such as silver sulfadiazine and burn dressing gauze (does not adhere to the burn) to prevent damage to the burned site and promote healing
- The child usually returns in 24 hours to assess their pain and observe site for signs of infection
- Blisters may be debrided (cut away) to remove possible necrotic tissue as the burn heals

EMERGENCY MANAGEMENT FOR SEVERE BURNS

- The child with a third- or fourth-degree burn is critically injured and requires immediate care
- Care includes fluid therapy, systemic antibiotic therapy, pain management, and physical therapy
- The goal is to prevent disability caused by scarring, infection, or contracture

EMERGENCY MANAGEMENT FOR ELECTRICAL BURNS OF MOUTH

- If a child puts the prongs of a plugged-in extension cord into the mouth or chews on an electric cord, the mouth can be burned severely
- When electrical current from a plug is conducted for a distance through the skin and underlying tissue, it can cause an ulcer
- If blood vessels were burned, active bleeding from the lesion will be present
- The immediate treatment for electrical burns of the mouth is to unplug the electric cord and control bleeding if present
- This can be done by pressing a towel against the burn site. In the emergency room, pain management and wound care are essential
- Clean the wound with an antiseptic solution

EMERGENCY MANAGEMENT FOR ELECTRICAL BURNS OF MOUTH

- Most children with electric burns are admitted to an observation unit for at least 24 hours because edema in the mouth could lead to airway obstruction
- Eating will be a challenge for the next week because the child's mouth is sore
- Soft foods and fluids may be easiest to swallow
- Electrical burns of the mouth turn black as local tissue necrosis begins
- They heal with white, fibrous scar tissue, possibly leaving a malformation of the lips or cheeks and difficulty speaking clearly afterward
- This can be minimized by using a mouth appliance, which helps maintain lip contour
- Many children need follow-up care by a plastic surgeon to restore the lip contour

THERAPY FOR BURNS

TABLE 52.4 COMPARING OPEN AND CLOSED BURN THERAPY

Method	Description	Advantages	Disadvantages
Open	Burn is exposed to air; used for superficial burns or body parts that are prone to infection, such as perineum	Allows frequent inspection of site; allows child to follow healing process	Requires strict isolation to prevent infection; area may scrape and bleed easily and impede healing
Closed	Burn is covered with antibiotic cream and nonadherent gauze; used for moderate and severe burns	Provides better protection from injury; is easier to turn and position child; allows child more freedom to play	Requires dressing changes that can be painful; possibility of infection is still present because of dark, moist environment

TOPICAL THERAPY FOR BURNS

- Silver sulfadiazine (Silvadene) is the drug of choice for burn therapy to limit infection at the burn site
- If Pseudomonas is detected in cultures, nitrofurazone (Furacin) cream may be applied

ESCHAROTOMY

- As natural protection for a burned area, a rigid scab (an eschar) forms over moderately or severely burned areas
- Fluid accumulates rapidly under an eschar, putting pressure on underlying blood vessels and nerves
- If an extremity or the trunk has been burned so that both anterior and posterior surfaces have eschar formation, a tight band may form around the extremity or trunk, cutting off circulation to distal body portions
- If distal parts feel cool to the touch and appear pale, any tingling or numbness is present, pulses are difficult to palpate, and capillary refill is slow (longer than 5 seconds), an escharotomy (cut into the eschar) may be performed
- Some bleeding will occur after escharotomy. Packing the wound and applying pressure usually relieves this

DEBRIDEMENT

- Debridement is the removal of necrotic tissue on which microorganisms could thrive from a burned area to reduce the possibility of infection.
- This may be done using collagenase (Santyl), an enzyme that dissolves devitalized tissue, or manually
- For manual debridement, children may have 20 minutes of hydrotherapy beforehand to soften and loosen eschar, which then can be gently removed with forceps and scissors
- Debridement is painful, and some bleeding occurs with it. Premedicate the child with a prescribed analgesic and help the child use a distraction technique during the procedure to reduce the level of pain.
- Transcutaneous electrical nerve stimulation (TENS) therapy or patient-controlled analgesia also can be helpful pain management measures.
- Extensive debridement is done using conscious sedation
- Praise any degree of cooperation. Plan an enjoyable activity afterward to aid in pain relief and to help reestablish some sense of control over the situation

GRAFTING

- Allografting: the placement of skin (sterilized and frozen) from cadavers or a donor on the cleaned burn site
  - Do not grow but provide temporary protective covering for area
- Autografting: process in which a layer of skin of both epidermis and a part of the dermis (called a split-thickness graft) is removed from a distal, unburned portion of the child's body and placed over the prepared burn site, where it will grow and replace the burned skin

NURSING CARE FOR BURNS

- Social isolation related to infection control precautions necessary to control spread of microorganisms
- Altered body image perception related to changes in physical appearance with burn injury